

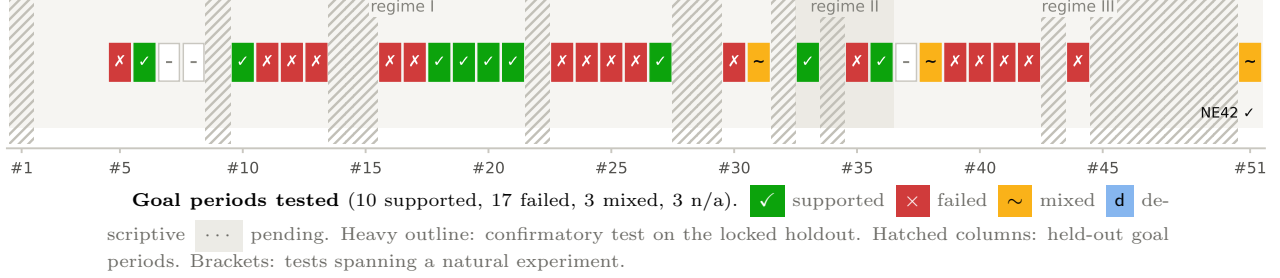
H41: Read-out gating gives the swarm a light cone

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

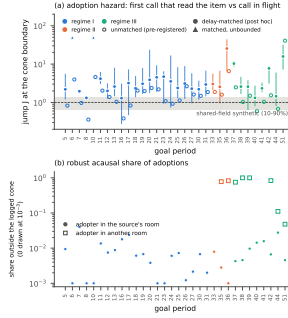
Bottom line. Novel items obey a logged light cone inside rooms and across the NE42 merge and split, but the pre-registered gating test mostly failed (it passes only after a post-hoc recency match), and cross-room leaks cannot be traced to any logged channel.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Can a novel item (a link, number, name or rare word) reach an agent before that agent's model calls could have read it? We traced 134k novel marker items (H34's rule; 45k adoptions) through the DQ1 context ledger in 32 non-holdout periods, building each item's logged light cone: the time-respecting read-out paths from its first use. Within a room the bound holds: adoptions outside the cone are rare (robust share 0–4% per period, median 0.6%). The pre-registered test of gating, the hazard at the first call that read the item over the call already in flight, passed in 12/32 periods. Matched for time since the item appeared (post hoc), it passes in 22/32. Across rooms the logged cone is a cage: 75–100% of cross-room adoptions lie outside it (7/8 two-room periods), and NE42's merge raised the cross-group hazard 80–860×, falling back at the split. The front is paced by talk turns, not cadence. Holdout untouched.



Observables. (a) Jump J per period: open circles are the pre-registered unmatched estimate (often ≤ 1 because adoption hazard decays with time since t_0 , and in-flight calls all sit near t_0); filled are delay-matched (post hoc). Gray: shared-field synthetic. (b) Share of adoptions outside the logged cone: adopters in the source's room (dots, $\leq 4\%$) versus another room (squares, 0.05–1).

PHYSICAL MODEL

Asynchronous kinetic updates (model 02) with contagion on a temporal graph (03). Agents update only at their model calls; a call reads the room messages posted since the previous call (the ledger). An item posted by source s at t_0 can reach agent i only along a time-respecting read-out path; K_i is i 's entry call into this logged cone. Adoption hazard per talk call: a shared field ε , unlogged channels η , plus q once i has read the item. J compares the hazard at the first entry call with the call in flight at t_0 .

$$T_s = t_0, \quad T_i = \min\{t_c^{\text{call}} : c \text{ reads } m, t_m \geq T_{\text{src}(m)}\}$$

$$\lambda_i(c) = \varepsilon(t_c) + \eta_i(c) + q\mathbb{I}[c \geq K_i^{\text{item}}]$$

$$\text{acausal: } c_{\text{use}} < K_i, \quad J = \frac{h(c = K_i)}{h(c = K_i - 1)} \Big|_{\text{same } t-t_0}$$

$$v = 1/\text{median}(\text{calls from exposure to use})$$

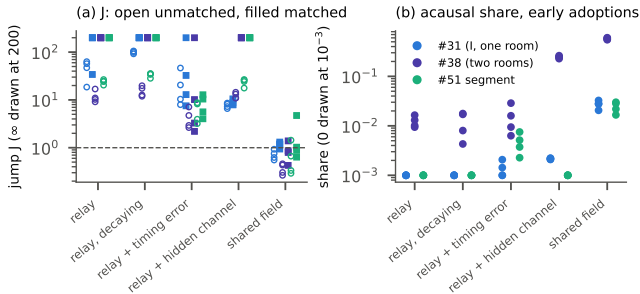
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator the useful part is negative: you cannot trace how cross-room information leaked from chat logs. Nearly all cross-room adoptions bypass the logged chat channel, yet shared repos and websites are as common among in-cone adoptions as among the leaks, so they explain nothing. Only humans re-posting an item in several rooms stand out (27× enriched). Rooms work as cages for chat: merging two groups multiplies cross-group pickup about a hundredfold. Inside a room, gating is real but weak: about a third of fast adoptions are co-generated (a shared field), not read. Spread runs at one hop per 1.5–5 talk turns, so tuning call cadence buys nothing.

H41-readout-light-cone · generated 2026-10-04 10:55 by infra/summaries/build_summaries.py · card hypotheses/H41-readout-light-cone/README.md · ratings: H41 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 J_{in} CI > 1 ($\geq 70\%$ II/III, $\geq 50\%$ I)	3/10 II/III, 8/19 I; matched (post hoc) 7/11, 15/21	fail (post hoc pass)
P2 $A_{rob} \leq 0.05$ (II/III), 0.10 (I)	29/32; the 3 misses are cross-room; within-room max 0.04	mostly
P3 early A below time-shuffled null	32/32, but a field also passes (synthetic)	non-specific
P4 cross-room adopters unreachable on pre-item graph	$\geq 90\%$ in 5/8 two-room periods; #35/36 bridged statically	partial
P5 cycles per hop: I ≤ 5 , I 47 / 5 talk; III 16 / 1.5 talk	III yes, I no	
P6 cadence $b \in [0.5, 1.5]$, $c > -0.2$	$b > 0$ (CI) in 3/25; c often < 0 or > 0	fail
P7 $\geq 50\%$ of violations have a channel	13/16, but controls 0.82	non-specific
P8 in-flight $\leq$ entry/3	2/10 (matched 4/10)	fail
NE42 merge $\geq 3 \times$ both ways; cage	$\times 82$, $\times 860$; cross acausal 1.0 / 0.02 / 1.0	yes
G38 cage, artifact leak	cage yes (ratio 0.001); artifact/web 0.19 vs 0.31	mixed



Second figure. Synthetic validation on real call schedules (#31, #38, #51; 4 replicates of 400 items). (a) The matched J separates relay (∞) from a shared field (≈ 1); timing error attenuates it but keeps the sign. (b) The acausal share separates field from relay only where logged paths are long (two rooms).

GOAL PERIODS IN DETAIL

G05 $\times$ failed
G06 $\checkmark$ supported
G10 $\checkmark$ supported
G11 $\times$ failed
G12 $\times$ failed
G13 $\times$ failed
G16 $\times$ failed
G17 $\times$ failed
G18 $\checkmark$ supported
G19 $\checkmark$ supported

+23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The headline gating result rests on a post-hoc estimator, chosen after the first real period and validated afterwards (8% false positives under a field). Items are hashed markers, so co-generated names and numbers look like adoptions. Channel attribution is non-specific because shared work is ubiquitous. Visibility is the ledger's room rule; its timing is estimated for non-Gemini agents. The cadence test lacks leverage: regime-III call intervals barely vary. NE42 is confounded with #40's shared goal. The first build mis-stored cone hop counts (fixed; all periods rebuilt). No multiplicity correction across 32 periods. Holdout not run.

ROUND 2

Pre-register the matched jump for the holdout. Time-stamp the unlogged channel with file-level git reads (DQ4) and web visits against matched controls. Test HH187's cycle-quantized lags on hop events. Use pause timers as exogenous cadence shifts.

- **Essence.** which channels actually move information between agents, and how fast. The logged chat channel is gated at the call scale and bounded by rooms, but a third of fast adoptions and almost all cross-room spread come from elsewhere.
- **H41-R1** Pre-register the matched jump and the class split.
- **H41-R2** Time-stamp the unlogged channel.
- **H41-R3** HH187: lags quantized in calls.
- **H41-R4** Co-generation as a field.
- **H41-R5** Exogenous cadence.